Soccer Players Statistics

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Data Overview

Overview:
This dataset showcases the attacking, defending, passing, and dribbling statistics of players of every single position in European football in the top 5 leagues (England, France, Germany, Italy, Spain). It only ranges the 2019-2020 season.

Why I selected this Dataset:
I have been a soccer fan my entire life, and have mainly watched European football. Recently I have been active in online soccer communities where debates like “what is the best league” or “what is the best generation of footballers” are common. I want to use the knowledge I have gained from this class so far and this dataset to gain some actual analytical insight into these topics.

Data Source:
https://www.kaggle.com/datasets/kriegsmaschine/soccer-players-values-and-their-statistics?utm_source=chatgpt.com&select=transfermarkt_fbref_201920.csv

Data Overview Continued

Data Distribution:
The data has significant outliers in most visualizations. For example, the market value variable has many outliers which can be seen in slides 9 and 10. This is because those outliers are the best and most valuable players in the world like Kylian Mbappe, Neymar Jr, Lionel Messi, Kevin De Bruyne, and Mohamed Salah.

Are there any notable data quality issues:
While there are many nulls, these null values have not actually affected my analysis much. Unlike other programming languages, Tableau allows aggregate functions to be carried out despite null values. The only thing affecting my analysis are outliers, which as pointed out, there is a valuable reason for.

Surprising Relationships Amongst Variables:
One thing which will be outlined later in the analysis is the relationship between what skills make players valuable per position. Growing up I always assumed that the skills native to a players position would make them valuable. For example, goals for an attacker, passing for a midfielder, and defending for a defender. However through my analysis, I learned that the better an attacker is at defending, and the better a midfielder is at scoring, the more valuable they are, and my best guess for this, is that modern teams value versatility.

Variables in the Dataset

The dataset contains basic variables like Age, birth year, league, minutes per 90, nationality, passing ability, player, and position.

- It contains variables that showcase a players passing ability such as: progressive passes, passes into the final third, passes pressure, and through balls. I summed these into a variable called passing ability for clarity.
- It contains variables that showcase a players defending ability such as blocked shots, interceptions, tackles won, and ball recoveries. I summed these into a variable called defending ability.
- It contains variables that showcase a players attacking ability such as Xg Xa Per 90, and Goals Assists per 90. I summed these into a variable called attacking ability.
- Finally it contains variables such as carry progressive distance, dribbles completed percentage, and players dribbled past. For this one, I did research online, and realized that the best statistic to measure dribbling across all players is carry progressive distance (this is because the units differ significantly and adding defeats the purpose).

Finally I created a variable called total ability. I did this by normalizing defending, passing, attacking, and progressive distance carried ((value - mean)/(standard deviation)). I then summed these normalized variables into total ability. This allows users to directly compare leagues and age groups.

Questions

How has football evolved over time in terms of what skills and abilities are prioritized?

How does age affect a players ability, and henceforth their playing time?

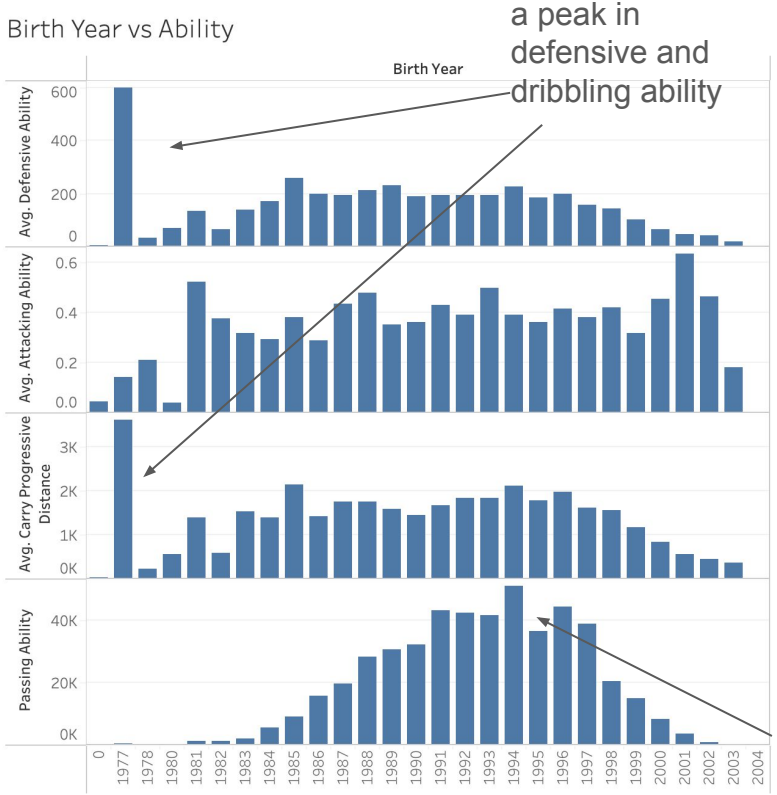
Does a players ability and their position have an affect on their market value?

What is the relationship between league and ability? (Are certain leagues better when it comes to an ability? Which leagues produce the best overall players?)

Visualizations

- The visualizations should answer at least 3 questions. You can choose to answer more if appropriate.
- Each visualization should have a title, clear axes, descriptive annotations, and a legend (as relevant)
- In addition, you should include a short caption (2-3 sentences) that describes the visualization and at least one key takeaway
- Map each visualization to the question(s) it addresses
- Your visualizations should be distinct from others. In other words, if 2 or more of your visualizations are too similar or overly simplistic, this may result in a deduction

How has football evolved over time in terms of what skills and abilities are prioritized?



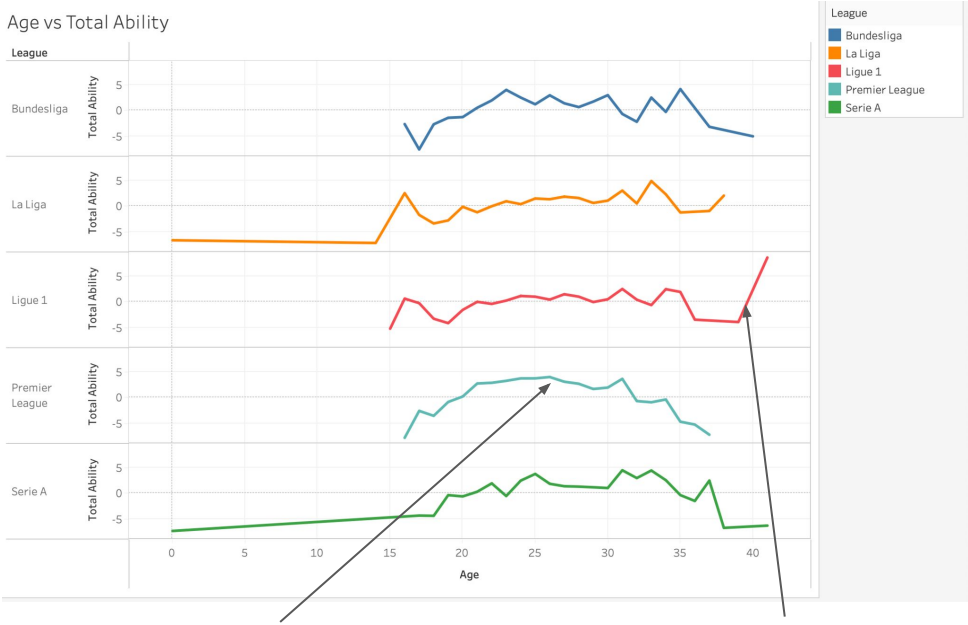
Here you can see a peak in defensive and dribbling ability

Here you can see a peak in passing ability, emphasizing the tactical shift

Caption: Through this visualization we can see that the defensive ability and dribbling ability of a footballer was prioritized more in earlier generations. A players attacking ability has always been constant through the generations, and passing was prioritized the most for the generation of players born in the early 90s.

Takeaway: The context of this dataset matters when analyzing these results. This is data collected in 2020. This means that anyone born around 2000 was still making their debuts (barely have any playing data). What this means is that attacking stayed consistent, defending and dribbling decreased, and passing grew in importance. This showcases the tactical shift in football. With teams prioritizing tactical movement of the player rather than physical movement of the players.

How does age affect a players ability, and henceforth their playing time? (ability)



You can see a beginning in the drop in quality in the Premier League

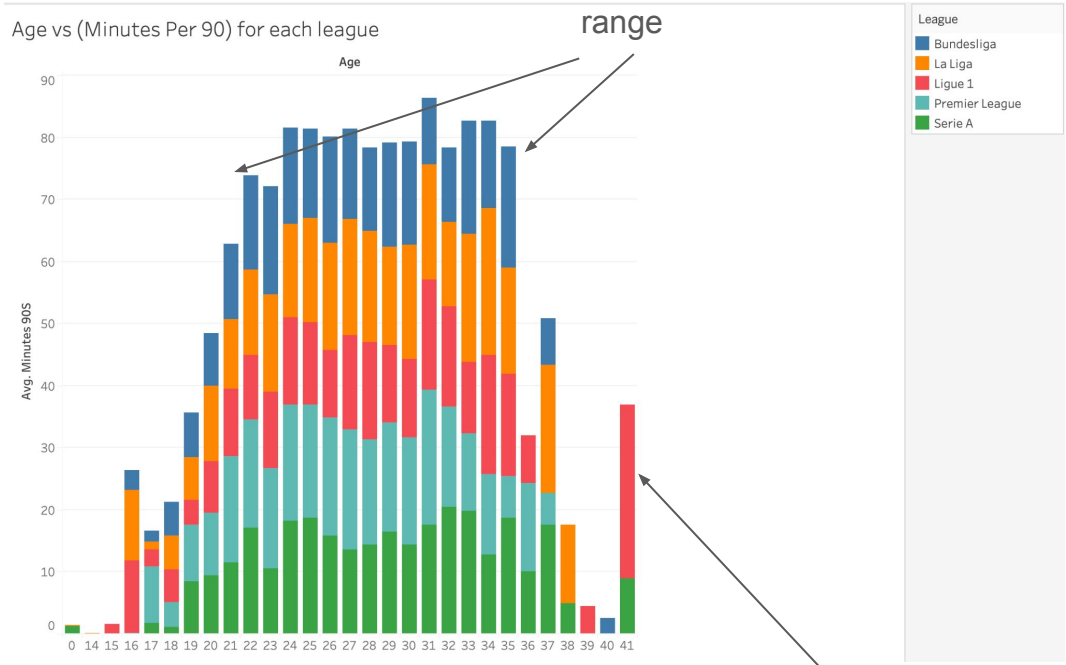
You can see an increase in quality in Ligue 1

Caption: This visualization showcases the total ability of a player over their playing lifetime (here age is shown as continuous) across each league. For almost every league the ability goes down at the end except for Ligue 1. The Premier League has the steepest decline at the end, and starts declining the earliest.

Takeaway: This showcases the difficult of the Premier League and how as you reach 28 to 30 years old, your ability (quality in this context) begins to decline. This showcases that the Premier League is one of the most difficult leagues compared to leagues like Ligue 1.

How does age affect a players ability, and henceforth their playing time? (playing time)

This is the max range

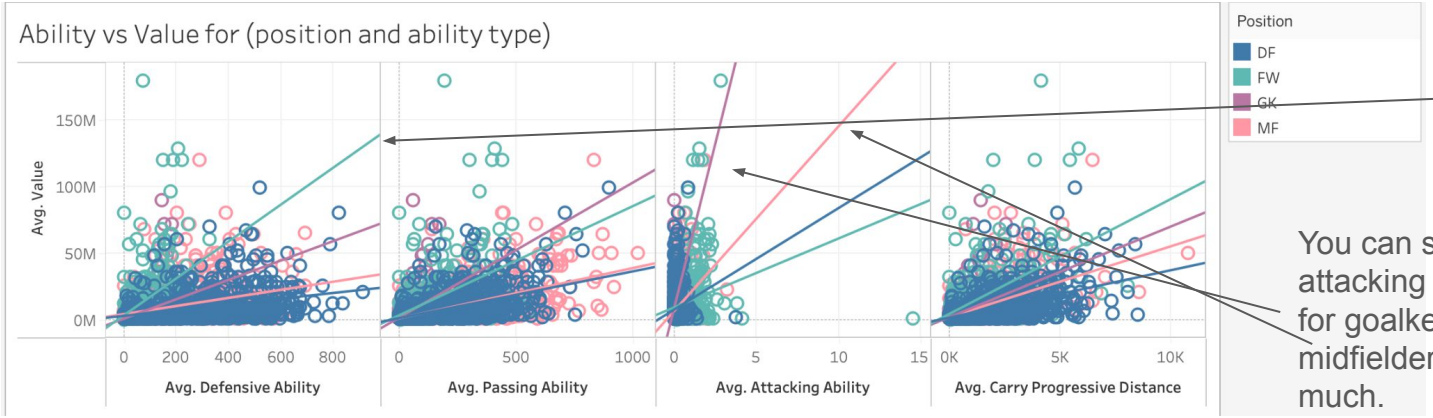


Here you can see the increase in Ligue1

Caption: This visualization shows the age vs minutes per 90 across each league. This data resembles a bell curve where the peak is in the middle. It also showcases the proportions of which league plays the most of each age. While the data is varying per league, one can notice that Ligue 1 has a smaller minutes per 90 average up until its later years. The peak of the bell curve can also be seen to be around 22 to 35 years, with the max reaching a range of 80 to 85 minutes.

Takeaway: This visualization shows that a player will reach their max playing time from the ages of 23 to 35, which will be around 85 minutes. They can also note that Ligue 1 isn;t the most ideal place to play until later in their career.

Does a players ability and their position have an affect on their market value? (ability)



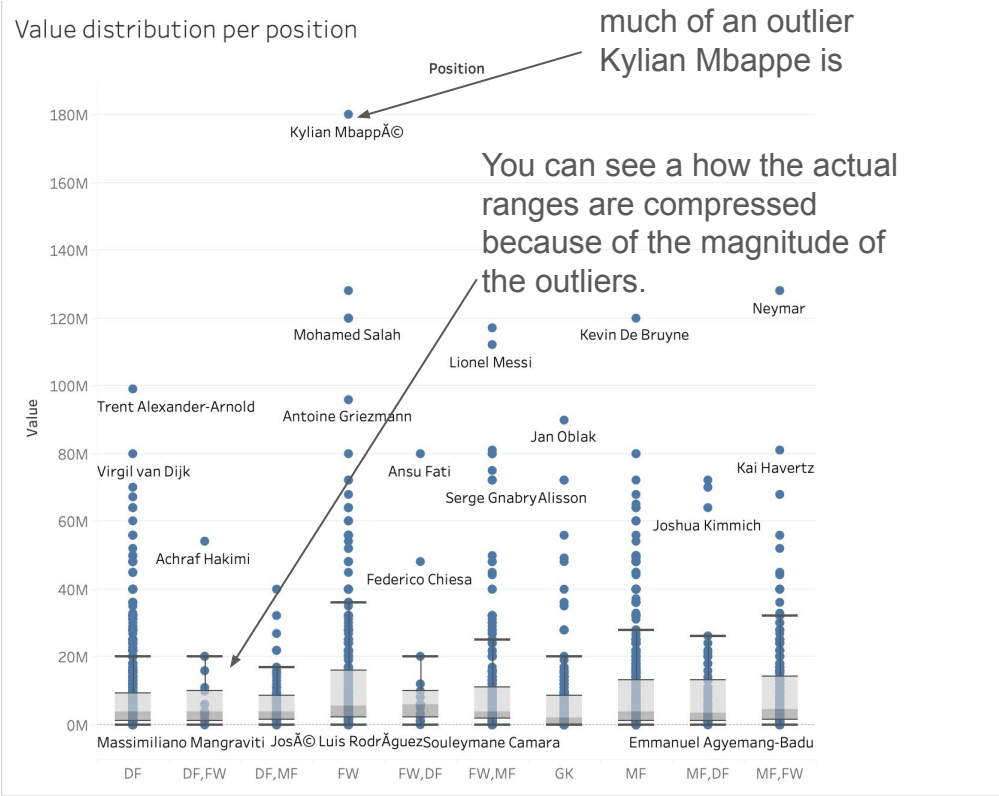
You can see that defending is important for attackers very much.

You can see that attacking is important for goalkeepers and midfielders very much.

Caption: This visualization shows the correlation between each ability and market value, per position. This visualization reveals something strange. Growing up I always assumed that the skills native to a players position would make them valuable. For example, goals for an attacker, passing for a midfielder, and defending for a defender. However, here you can see that defending and dribbling is best for an attacker when it comes to optimizing their value, and attacking ability is best when it comes to optimizing a midfielder and goalkeepers value.

Takeaway: While this may seem very odd, an underlying reason for this can be a modern day emphasis on versatility. The more versatile your team is, the better your team is when it comes to defending together, attacking together, and passing together. After all, football is a team sport.

Does a players ability and their position have an affect on their market value? (position)



You can see a how much of an outlier Kylian Mbappe is

You can see a how the actual ranges are compressed because of the magnitude of the outliers.

Caption: This visualization shows the value of each footballer based on position. By no surprise to any football fan, attackers, midfielders, and forwards are the highest in value. However on interesting thing this visualization shows is the disparity between the top 1% and the rest of the world. Every dot is an outlier. If you look at Kylian Mbappe, not only is he an outlier, but the distance between him and the actual range of data shrinks the range down to an indecipherable point.

Takeaway: This showcases that even despite the vast number of footballers in the world, the elite few that we see play regularly are statistically speaking, outliers.

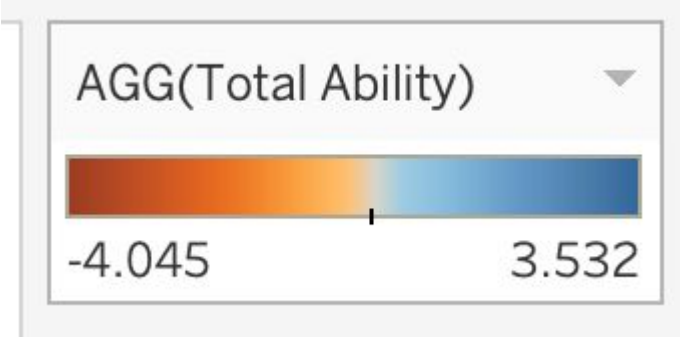
What is the relationship between league and ability? (Are certain leagues better when it comes to an ability? Which leagues produce the best overall players?)

League vs Total Ability

League	
Bundesliga	1.298
La Liga	-1.252
Ligue 1	-4.045
Premier League	3.532
Serie A	0.466

Ligue 1 is very far behind

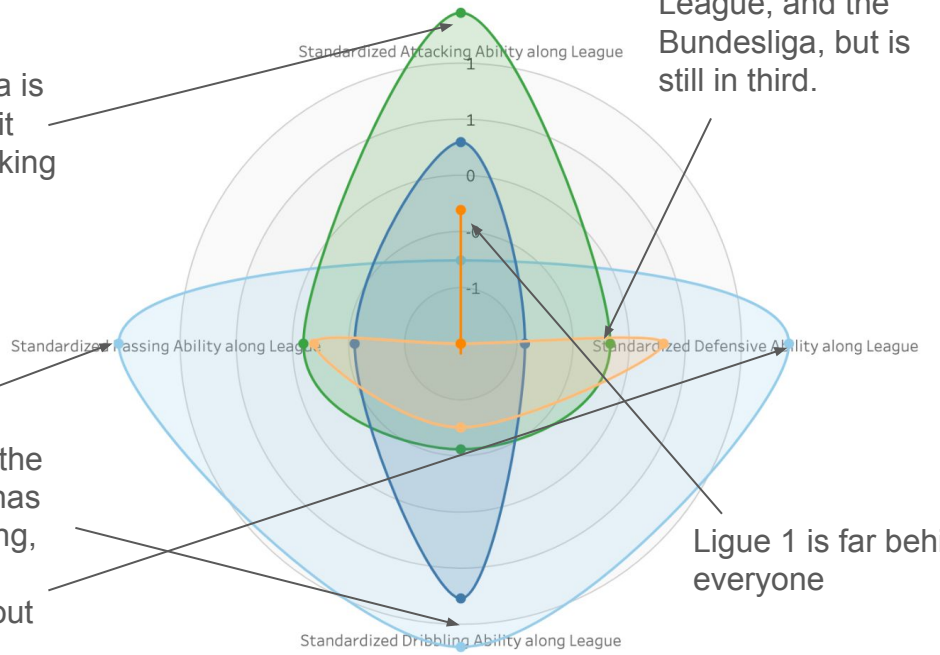
The Premier League has the highest ability, showing its quality



Caption: This visualization shows the average total ability of each league in the top 5 leagues. From this visualization, one can see that the Premier League is the highest, compared to Ligue 1 which has the lowest average ability.
Takeaway: From this visualization we can clearly see that certain leagues are better when it comes to overall total ability (passing, attacking, dribbling and defending).

What is the relationship between league and ability? (Are certain leagues better when it comes to an ability? Which leagues produce the best overall players?)

Standardized Ability Types across Leagues



The Bundesliga is the best when it comes to attacking

La Liga is behind both the Premier League, and the Bundesliga, but is still in third.

You can see that the Premier League has the highest passing, dribbling, and defensive ability out of every league

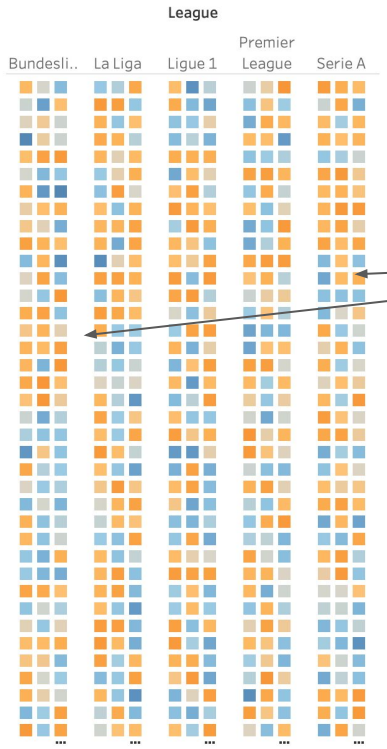
Ligue 1 is far behind everyone

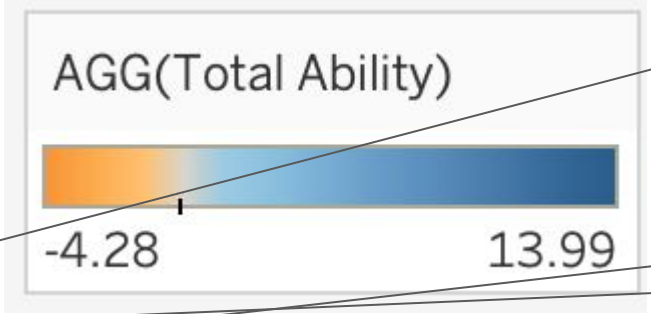
Caption: From this visualization we can see the specific areas in which teams differ. Here, one can see that the Premier is more talented than the others when it comes to everything except attacking, while that is the area in which the Bundesliga excels.

Takeaway: Not only can we clearly see which teams are better, but this also is in accordance with our previous visualization showing that the Premier League and Bundesliga are the highest in overall quality.

What is the relationship between league and ability? (Are certain leagues better when it comes to an ability? Which leagues produce the best overall players?)

Total Ability per Player divided by League





You can see that the variation in color between leagues is very minimal

Most boxes are orange indicating most players have a lower total ability.

Caption: This visualization complements the previous 2 visualizations showcasing the overall ability for each individual player split up by league. Here we can see that the difference in quality is not extremely visible like the previous 2 visualizations.

Takeaway: This observation pointed out in the caption is crucial. Just because the overall quality of one league is better than another, doesn't mean that almost every single player is better. The difference in quality lies in the difference between attributes in every player when considering the calculations done to compute the average abilities.

Reflection

What did you struggle with? And what did you enjoy? (~2-3 sentences)

One huge thing I struggled with was figuring out how to make at least 8 different visualizations for only 1 dataset. I found myself repeating visualizations (in the sense of their purpose) multiple times. However, after concluding this process and finishing my visualizations I truly learned what makes certain visualizations valuable. For example when debating where a line graph is best used I realized if age were a continuous value, a line graph could showcase so many things like ability or value over time. I enjoyed this specific project very much because I finally found a way to analyze the game I love with data science. Gaining a true insight into which players and leagues are the best informed me.

What did you learn? (~3-4 sentences)

I learnt that footballers need to be more versatile in the modern game. Furthermore, they need to prioritize passing as the game has experienced a tactical shift. From a data science perspective, I learned that standardization/normalization is a valuable tool that can allow us to efficiently compress data.